Egg Drops: Comparing the 1–3 and 2–2 Strategies

September 11, 2026

Main conclusion. For large N, the 1–3 strategy needs

$$(24N)^{1/4} + O(1) = 24^{1/4} N^{1/4} + O(1) \approx 2.21336\, N^{1/4},$$

while the best fixed-block 2–2 strategy needs

$$2\sqrt{2}\, N^{1/4} + O(1) \approx 2.82843\, N^{1/4} + O(1).$$

Thus the 1–3 strategy is asymptotically better. Its leading constant is about 21.75% smaller, or equivalently the 2–2 strategy uses about 27.79% more drops in the leading-order comparison. In fact, the 1–3 strategy is an optimal four-egg strategy, so no restricted 2–2 construction can beat it for any finite N when both are interpreted as worst-case drop counts.

1 The basic egg-drop counting function

Suppose we have k eggs and t drops available. Let

$$F_k(t) = \text{maximum number of floors for which we can always locate the critical floor.}$$

Here "critical floor" means the highest floor that is safe, with the usual monotonic rule: if an egg breaks at a floor, it also breaks above that floor; if it survives, it survives below that floor.

For $t = 0$, we have $F_k(0) = 0$. For $t \geq 1$, make a first drop. There are two cases.

- The egg breaks. Then we have $k - 1$ eggs and $t - 1$ drops remaining, so we can handle $F_{k-1}(t-1)$ floors below the test floor.
- The egg survives. Then we still have k eggs and $t - 1$ drops, so we can handle $F_k(t-1)$ floors above the test floor.

Therefore

$$F_k(t) = F_{k-1}(t-1) + 1 + F_k(t-1). \tag{1}$$

The $+1$ is the floor on which the first egg is dropped.

Iterating the recurrence gives the familiar closed form

$$F_k(t) = \sum_{j=1}^{k} \binom{t}{j}. \tag{2}$$

For example,

$$F_2(t) = \binom{t}{1} + \binom{t}{2} = \frac{t(t+1)}{2},$$

$$F_3(t) = \binom{t}{1} + \binom{t}{2} + \binom{t}{3},$$

and

$$F_4(t) = \binom{t}{1} + \binom{t}{2} + \binom{t}{3} + \binom{t}{4}.$$

Let

$$T_k(N) = \min\{t : F_k(t) \geq N\}$$

be the least worst-case number of drops needed with k eggs and N floors.

2 The 1–3 strategy

The phrase "1–3" means that we use one egg for the outer search and reserve the other three eggs for the lower-level search after the first egg breaks. But this is exactly the recursive strategy encoded in (1) for $k = 4$.

Suppose we want a four-egg strategy using t drops. We should place the first egg at a height such that, if it breaks, the remaining three eggs and $t - 1$ drops can resolve every possible lower critical floor. Since that lower part can contain at most $F_3(t - 1)$ floors, we should put the first egg at floor

$$F_3(t - 1) + 1.$$

If it breaks, solve the lower part with the three remaining eggs. If it survives, continue upward with all four eggs and $t - 1$ drops. Thus the total number of floors covered is

$$F_3(t - 1) + 1 + F_4(t - 1) = F_4(t).$$

Consequently, the exact worst-case drop count for the 1–3 strategy is

$$T_{13}(N) = T_4(N), \qquad T_4(N) = \min\left\{ t : \sum_{j=1}^{4} \binom{t}{j} \geq N \right\}. \tag{3}$$

This observation is important: the 1–3 strategy is not merely one reasonable four-egg strategy. It is the standard optimal four-egg recursion. Therefore any other four-egg strategy, including a 2–2 construction, cannot have a smaller exact worst-case drop count.

Asymptotics

Expand $F_4(t)$:

$$\begin{aligned} F_4(t) &= t + \frac{t(t-1)}{2} + \frac{t(t-1)(t-2)}{6} + \frac{t(t-1)(t-2)(t-3)}{24} \\ &= \frac{t^4}{24} + \frac{t^3}{12} + \frac{11t^2}{24} + t. \end{aligned}$$

The leading term is $t^4/24$. Hence the equation $F_4(t) \approx N$ gives

$$\frac{t^4}{24} \approx N,$$

so

$$T_{13}(N) = 24^{1/4} N^{1/4} + O(1). \tag{4}$$

Numerically,

$$24^{1/4} \approx 2.21336384.$$

3 The 2–2 strategy

Now follow the proposed block strategy. Fix a positive integer F. Group the floors into superfloors of size F:

$$\{1, \ldots, F\}, \quad \{F+1, \ldots, 2F\}, \quad \ldots$$

There are

$$M = \left\lceil \frac{N}{F} \right\rceil$$

superfloors.

In the first stage we use two eggs to determine which superfloor contains the critical floor. In the second stage, once the critical superfloor is known, we use two eggs to search through at most F ordinary floors.

For two eggs,

$$F_2(t) = \frac{t(t+1)}{2}.$$

Therefore

$$T_2(m) = \min\left\{ t : \frac{t(t+1)}{2} \geq m \right\} = \left\lceil \frac{\sqrt{8m+1} - 1}{2} \right\rceil. \tag{5}$$

Thus the exact drop count of the fixed-F 2–2 strategy is

$$T_{22}(N; F) = T_2\left(\left\lceil \frac{N}{F} \right\rceil \right) + T_2(F), \tag{6}$$

and the best such strategy is

$$T_{22}(N) = \min_{1 \leq F \leq N} \left[T_2\left(\left\lceil \frac{N}{F} \right\rceil \right) + T_2(F) \right]. \tag{7}$$

4 Finding the right block size asymptotically

The exact formula (7) is an integer optimization problem. To see its scale, use

$$T_2(x) = \sqrt{2x} + O(1),$$

which follows immediately from (5). Ignoring the ceiling and lower-order integer effects for a moment gives

$$T_{22}(N;F) \approx \sqrt{\frac{2N}{F}} + \sqrt{2F}.$$

Write

$$f(F) = \sqrt{\frac{2N}{F}} + \sqrt{2F}.$$

Differentiate with respect to real $F > 0$:

$$f'(F) = -\frac{\sqrt{2N}}{2F^{3/2}} + \frac{\sqrt{2}}{2F^{1/2}}.$$

Set $f'(F) = 0$:

$$-\frac{\sqrt{2N}}{2F^{3/2}} + \frac{\sqrt{2}}{2F^{1/2}} = 0.$$

Multiplying by $2F^{3/2}/\sqrt{2}$ gives

$$-\sqrt{N} + F = 0,$$

so the continuous optimum occurs at

$$F \sim \sqrt{N}. \tag{8}$$

There is an even cleaner way to see the same answer. By AM–GM,

$$\begin{aligned}\sqrt{\frac{2N}{F}} + \sqrt{2F} &\geq 2\sqrt{\sqrt{\frac{2N}{F}}\sqrt{2F}} \\ &= 2\sqrt{2}\, N^{1/4},\end{aligned}$$

with equality exactly when $F = \sqrt{N}$.

Therefore the leading asymptotic cost of the optimized 2–2 construction is

$$T_{22}(N) = 2\sqrt{2}\, N^{1/4} + O(1) \approx 2.82842712\, N^{1/4} + O(1). \tag{9}$$

Remark 1. *For finite N, one should not blindly set $F = \sqrt{N}$. The exact optimum is an integer F minimizing (7); the triangular-number ceilings can shift the best F noticeably. What remains true is that every asymptotically optimal choice has F on the order of $\sqrt{N}$, and the leading constant is $2\sqrt{2}$.*

5 Which constant wins?

We now compare the leading terms:

$$\boxed{\text{1–3: } 24^{1/4} \approx 2.21336}$$

and

$$\boxed{\text{2–2: } 2\sqrt{2} \approx 2.82843}.$$

Since

$$2.21336 < 2.82843,$$

the 1–3 strategy wins asymptotically.

The ratio is

$$\frac{2\sqrt{2}}{24^{1/4}} \approx 1.27788621. \tag{10}$$

Thus, at the level of leading asymptotic constants, the 2–2 strategy takes about 27.79% more drops than 1–3:

$$1.27788621 - 1 \approx 0.27788621.$$

Equivalently, the 1–3 strategy reduces the leading drop count by

$$1 - \frac{24^{1/4}}{2\sqrt{2}} \approx 0.21746,$$

about 21.75% relative to the 2–2 count.

6 Why the constants have such different shapes

It is helpful to understand where the constants come from before doing any arithmetic.

With four eggs, the exact capacity after t drops is

$$F_4(t) = \binom{t}{1} + \binom{t}{2} + \binom{t}{3} + \binom{t}{4}.$$

The highest-order term is $\binom{t}{4} \sim t^4/4!$. Hence

$$N \sim \frac{t^4}{4!}, \qquad t \sim (4!N)^{1/4} = 24^{1/4} N^{1/4}.$$

The 2–2 construction throws away some of that four-dimensional efficiency by forcing the search into two separate square-root stages. Each two-egg stage costs about the square root of its search space:

$$\text{top stage} \sim \sqrt{\frac{2N}{F}}, \qquad \text{bottom stage} \sim \sqrt{2F}.$$

Balancing those two terms forces $F \sim \sqrt{N}$ and produces

$$2\sqrt{2}\, N^{1/4}.$$

The exponent is still $1/4$, but the decomposition has a larger leading constant.

This is a useful general lesson: once two algorithms have the same power of N, one must keep the leading constants. Here the difference is substantial, not a small rounding effect.

7 A few concrete values

The exact formulas make the finite-N comparison transparent. The following table uses

$$T_{13}(N) = T_4(N)$$

and minimizes (6) over integer F.

N	$T_{13}(N)$	best $T_{22}(N)$	an optimal F
100	8	8	10
1,000	13	15	28
10,000	23	28	66
1,000,000	71	89	817

The $N = 100$ tie is a good reminder that asymptotics do not mean that the winner must be separated for every small input. But the exact four-egg optimum is always no worse than the restricted 2–2 family, and the asymptotic gap eventually becomes large in absolute terms.

For example, at $N = 10^6$ the leading-order estimates are

$$24^{1/4}(10^6)^{1/4} \approx 69.99, \qquad 2\sqrt{2}(10^6)^{1/4} \approx 89.44,$$

which are already close to the exact values 71 and 89.

8 Final answer

Proposition 1. *For the egg-drop problem with 4 eggs and N floors, the 1–3 strategy has worst-case cost*

$$T_{13}(N) = \min\left\{ t : \sum_{j=1}^{4} \binom{t}{j} \geq N \right\} = 24^{1/4} N^{1/4} + O(1).$$

For the fixed-block 2–2 strategy,

$$T_{22}(N) = \min_{F \in \{1,\ldots,N\}} \left[T_2\left(\left\lceil \frac{N}{F} \right\rceil \right) + T_2(F) \right] = 2\sqrt{2}\, N^{1/4} + O(1).$$

Hence

$$24^{1/4} < 2\sqrt{2},$$

so 1–3 is the better strategy. Its leading constant is about 21.75% smaller than the 2–2 leading constant.

The conceptual reason is simple: the 1–3 recursion uses all four eggs in an integrated way, and the binomial coefficient $\binom{t}{4}$ gives the $1/4!$ factor that determines the constant. The 2–2 method deliberately separates the problem into two two-egg searches, and that separation raises the constant even though it preserves the same $N^{1/4}$ exponent.

Sources and further reading

Steven J. Miller's mathematics talks page lists the Hampshire College talk "Egg Drop Mathematics: It's all it's cracked up to be," given July 28, 2026, together with the associated video. See https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/talks.html

Xiangwen Cao, Zongyun Chen, and Steven J. Miller, "Egg Drop Problems: They Are All They Are Cracked Up To Be!," *The Fibonacci Quarterly*. The paper is available from the authors' Williams

College site and on arXiv (arXiv:2511.18330): https://arxiv.org/abs/2511.18330

The talk and video supplied with the question: https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/eggdropmath_Hampshire28Jul2026.pdf
https://youtu.be/98cpdt2EsCM

The paper URL supplied with the question: https://web.williams.edu/Mathematics/sjmiller/public_html/math/papers/EggDropProblems_FibonacciQuarterly.pdf